

Semantics-Guided Feature Cutting and Batch Calibration for Low-Resource Multimodal Depression Assessment

Zhengshuo Gong
School of Artificial Intelligence,
Beijing University of Posts and
Telecommunications
Beijing, China
gongzs729@bupt.edu.cn

Ziyi Guo
School of Artificial Intelligence,
Beijing University of Posts and
Telecommunications
Beijing, China
guoziyi2023@bupt.edu.cn

Junyang Wu
School of Artificial Intelligence,
Beijing University of Posts and
Telecommunications
Beijing, China
wujunyang666@bupt.edu.cn

Yiming Gao
School of Artificial Intelligence,
Beijing University of Posts and
Telecommunications
Beijing, China
m15834150769@163.com

Zilu Zhang
School of Artificial Intelligence,
Beijing University of Posts and
Telecommunications
Beijing, China
zhangzilu@bupt.edu.cn

Yiming Jiang
School of Artificial Intelligence,
Beijing University of Posts and
Telecommunications
Beijing, China
jiang2005@bupt.edu.cn

Yingying Zhou
School of Artificial Intelligence,
Beijing University of Posts and
Telecommunications
Beijing, China
yingyingzhou@bupt.edu.cn

Yingming Gao^{*}
School of Artificial Intelligence,
Beijing University of Posts and
Telecommunications
Beijing, China
yingming.gao@bupt.edu.cn

Yiming Ding
School of Artificial Intelligence,
Beijing University of Posts and
Telecommunications
Beijing, China
lololodym@bupt.edu.cn

Liang Tan
School of Artificial Intelligence,
Beijing University of Posts and
Telecommunications
Beijing, China
2023211884@bupt.cn

Jingyao Xue
School of Artificial Intelligence,
Beijing University of Posts and
Telecommunications
Beijing, China
jingyaoxue@bupt.edu.cn

Ya Li^{*}
School of Artificial Intelligence,
Beijing University of Posts and
Telecommunications
Beijing, China
yli01@bupt.edu.cn

Abstract

The MPDD-AVG challenge studies a valuable problem in affective computing and mental-health assessment: estimating depression severity from multimodal behavioural signals. The central difficulty is low-resource supervision: using 88 labelled Young-track training subjects, a system must select useful cues from high-dimensional audio, video, gait, and personality data. We therefore build a compact challenge system around three design principles: semantics-guided feature cutting, shallow prediction, and metric-aware post-processing. The final pipeline uses a gait-personality ordinal-ridge backbone, four OpenFace AU/gaze temporal predictors, linear fusion, transductive batch-level moment alignment for CCC, and a backbone-confirmed classification rule. We also report exploratory routes across the Young modality settings: a G+P backbone, audio and ASR studies, and the final A-V-G+P submission family. Local analyses support compact IMU and OpenFace representations over several raw or higher-dimensional alternatives, while preserved public-leaderboard checkpoints document gains associated with facial AU/gaze dynamics, moment alignment, and conservative classification. The final system achieves a public leaderboard Score of **0.8745** on the MPDD-AVG 2026 Young track.

^{*}Corresponding authors.

CCS Concepts

• **Computing methodologies** → **Machine learning**; • **Applied computing** → *Health informatics*.

Keywords

multimodal depression detection, low-resource learning, facial behaviour analysis, gait analysis, concordance calibration, PHQ-9 prediction

1 Introduction

Depression is a major mental-health condition worldwide [21]. In clinical practice, screening tools such as PHQ-9 provide a structured way to measure symptom severity [13], while behavioural cues expressed through facial movement, gait, speech interaction, and personality-related traits can provide complementary information. The MPDD-AVG challenge offers a valuable benchmark for studying this problem under a unified multimodal protocol, requiring systems to estimate both continuous PHQ-9 severity and discrete depression labels.

A central difficulty of this setting is stable modelling under limited supervision. Depression-related multimodal data are inherently

difficult and sensitive to collect, while raw behavioural descriptors can be high-dimensional and noisy. For the Young track, this means that simply increasing model capacity is not necessarily the best strategy. Instead, the system must control variance, retain interpretable signals, and make modelling choices that are robust under small-sample training.

The work reported here is organized around the Young modality settings rather than only the best final score. The G+P setting motivates the stable gait–personality backbone. Audio and ASR routes are retained as non-final evidence about speech and language cues under limited supervision. The final A-V-G+P submission family then combines the G+P backbone with compact visual OpenFace predictors, while excluding speech branches that did not improve the recorded development results.

Our final system follows this principle. We build a compact pipeline that first cuts each modality to semantically meaningful features, then combines several lightweight predictors. The main backbone, denoted ORIG, uses the first three IMU acceleration channels together with Big-Five personality scores and fits an ordinal ridge regressor for PHQ-9 prediction. In parallel, the facial branch uses OpenFace AU and gaze streams only, summarised by four complementary temporal predictors: all-event statistics and facial change, event-3 state, cross-event drift, and spectral facial dynamics. The resulting PHQ-9 predictions are linearly fused, calibrated with transductive batch-level moment alignment for CCC, and converted into binary and ternary labels with a backbone-confirmed rule.

Contributions. This paper makes four contributions. (1) We present a fully specified low-resource system for MPDD-AVG Young, combining gait–personality modelling, OpenFace temporal predictors, batch calibration, and rule-based label derivation. (2) We organize the system history across the Young modality settings: G+P for the backbone, audio/ASR and deep sequence models as non-final explorations, and A-V-G+P for the final multimodal submission family. (3) We provide local evidence for semantics-guided feature cutting, where compact IMU and OpenFace representations outperform several raw or higher-dimensional alternatives in recorded analyses. (4) We separate controlled local design evidence from public-leaderboard development feedback, including the transductive calibration and final threshold choices. The submitted system achieves a public leaderboard Score of **0.8745** on the MPDD-AVG 2026 Young track.

2 Related Work

2.1 Multimodal Depression Detection

Automatic depression detection has been widely studied with interview, speech, facial, language, and physiological or behavioural signals. This fits the broader multimodal machine-learning setting, where complementary modalities must be represented, aligned, fused, and regularized under limited supervision [2]. Early benchmark resources such as DAIC-WoZ provided structured human–computer interview data for distress analysis [10], while AVEC standardised audio-visual depression and affect recognition evaluation protocols [18]. More recent datasets extend this direction to naturalistic or clinical multimodal settings, including D-Vlog [22], EATD [19], and MODMA [4]. The MPDD challenge introduces personality-aware

depression detection and provides a unified benchmark for multimodal PHQ-9 estimation [8]. Our work follows this multimodal line, but focuses on a compact system design rather than high-capacity end-to-end fusion.

2.2 Personality and Facial Behaviour Modelling

Personality information is relevant to depression assessment because individual traits can affect both symptom expression and behavioural responses. We use the Big-Five trait representation, a standard taxonomy for personality measurement [12]. Prior MPDD systems have therefore explored personality-enhanced multimodal fusion [20], hypergraph-based personality-aware modelling [9], and comparative machine-learning or language-model approaches [11]. Facial behaviour is another commonly used signal in depression detection, because reduced expressiveness, gaze changes, and action-unit dynamics can reflect affective and psychomotor states. AU terminology follows the Facial Action Coding System [6]. We use OpenFace [3] to extract interpretable facial streams, but retain only AU and gaze features rather than using the full raw descriptor. This choice connects the facial branch to explicit behavioural cues and keeps the representation compact.

2.3 Low-Resource and Metric-Aware Modelling

Mental-health datasets are naturally difficult to scale because of privacy, annotation cost, and participant recruitment constraints. Under such conditions, high-dimensional representations can introduce large estimation variance, making feature selection and task-driven dimensionality reduction important. Our system uses this low-resource bias throughout: IMU streams are reduced to acceleration-based gait statistics, personality descriptions are reduced to Big-Five scores, and OpenFace outputs are reduced to AU/gaze streams. We nevertheless explored speech representations such as openSMILE descriptors [7] and wav2vec2 embeddings [1]; these routes are reported as negative evidence rather than final-system components. For the facial predictors, we additionally use mixup [23] and seed averaging to reduce instability.

The MPDD-AVG score also encourages metric-aware modelling. Continuous PHQ-9 prediction is evaluated with the concordance correlation coefficient (CCC), which penalises not only rank disagreement but also mean and variance mismatch [14]. Discrete labels are assessed with classification and agreement metrics, including Cohen’s kappa [5]. These metrics differ from ordinary regression losses and motivate our transductive moment-alignment and rule-based classification steps. The backbone further treats PHQ-9 as an ordered severity target through ordinal ridge regression, implemented with the mord ordinal regression package [17]. These choices allow the final system to remain simple while matching the structure of the official evaluation.

3 Methodology

3.1 Feature Cutting

Figure 1 summarizes the submitted system. It is designed around a simple observation: in this task, the number of labelled training subjects is much smaller than the dimensionality of many raw behavioural descriptors. Directly fitting high-dimensional video or sensor descriptors can therefore produce unstable predictors

MPDD-AVG Young — System Pipeline

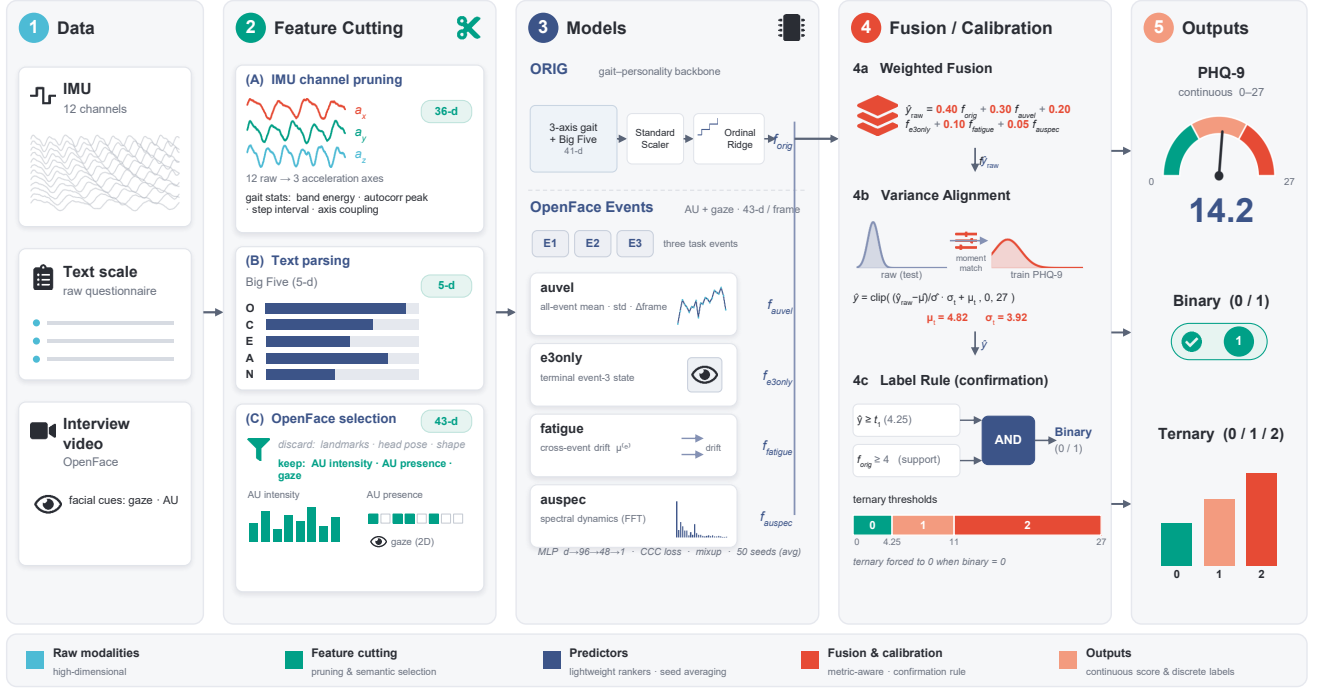


Figure 1: Overview of the submitted MPDD-AVG Young system. The pipeline first cuts raw IMU, personality-scale text, and OpenFace streams into compact behavioural cues, then applies an ordinal gait–personality backbone and four OpenFace temporal predictors. The five continuous predictions are linearly fused, batch-calibrated for CCC, and converted into binary and ternary labels with backbone confirmation.

whose apparent training signal is dominated by sample-specific correlations. We use feature cutting as the first modelling step: each retained modality is reduced to a small set of semantically interpretable signals before any supervised model is trained.

The final system keeps three information sources. First, from the IMU stream we retain only the first three channels, corresponding to the acceleration axes (a_x, a_y, a_z). These channels directly describe gait rhythm, movement intensity, and body-motion periodicity. The remaining IMU channels are not treated as physically meaningless, but they substantially increase the feature-to-sample ratio and were less reliable in our dimension-selection experiments. Second, from the personality descriptions we parse only the five Big-Five scores. This converts free-form text into a compact personality vector and avoids introducing a high-dimensional language embedding into the final model. Third, from OpenFace we discard landmarks, head pose, and shape parameters, and keep only facial behaviour streams: gaze, action-unit (AU) intensities, and AU presence indicators. This gives an interpretable 43-dimensional per-frame facial signal rather than a raw OpenFace descriptor.

This cutting step is not a post-hoc compression of a large trained model. It is part of the modelling assumption: semantics-guided feature selection and temporal summarisation are used to control estimation variance under limited supervision. The later predictors

are deliberately shallow relative to sequence-level end-to-end alternatives: an ordinal ridge backbone for IMU/personality and two-hidden-layer MLPs for facial temporal summaries. The goal is to preserve behavioural cues relevant to depression severity—gait dynamics, personality traits, gaze, and facial action patterns—without fitting the full raw descriptors directly.

3.2 ORIG Backbone

The backbone, denoted ORIG, uses the acceleration channels and Big-Five personality scores. Let s be one acceleration axis sampled at $f_s = 50$ Hz. For each axis, we extract band-energy, variation, periodicity, and peak-interval features. We first apply fourth-order Butterworth band-pass filters with zero-phase `filtfilt` filtering to obtain three frequency components: a gait band (0.5–1.5 Hz), a locomotion band (1.5–3 Hz), and a high-frequency band (3–10 Hz). For band b_k , the relative energy is

$$r_k = \frac{\sum_t s_{b_k}(t)^2}{\sum_t s(t)^2 + \epsilon}, \quad k \in \{1, 2, 3\}. \quad (1)$$

For the two lower-frequency bands, we also compute the coefficient-of-variation statistic used in the submitted code,

$$CV_k = \frac{\text{std}(s_{b_k})}{|\text{mean}(s_{b_k})| + \epsilon}. \quad (2)$$

with $\epsilon = 10^{-8}$. Because a band-pass output can have a small mean, this quantity should be interpreted as a code-level descriptive statistic rather than a physically calibrated variability measure; the downstream scaler standardizes all IMU features before regression.

To capture gait periodicity, we compute the normalized autocorrelation

$$R(\tau) = \frac{\sum_t (s_t - \bar{s})(s_{t+\tau} - \bar{s})}{\sum_t (s_t - \bar{s})^2}, \quad (3)$$

and record the first peak in the lag range $\tau \in [1, 100)$ whose height is at least 0.3, together with the corresponding period τ/f_s . We further detect signal peaks with a minimum distance of $f_s/4$ and prominence 0.5, then compute the mean, standard deviation, and coefficient of variation of inter-peak intervals. Thus each acceleration axis contributes 10 features: three band-energy ratios, two band CVs, autocorrelation peak height, dominant period, and three peak-interval statistics.

Across the three axes we add six aggregate features: the pairwise correlations ρ_{xy} , ρ_{xz} , ρ_{yz} and the mean, standard deviation, and coefficient of variation of acceleration magnitude

$$\|a\| = \sqrt{a_x^2 + a_y^2 + a_z^2}. \quad (4)$$

The IMU representation is therefore $3 \times 10 + 6 = 36$ dimensional. We concatenate five parsed Big-Five scores, yielding a 41-dimensional input vector for the backbone.

All backbone features are standardized with training-set statistics:

$$\tilde{x}_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j + \epsilon}, \quad \mu_j, \sigma_j \text{ estimated on the training split only.} \quad (5)$$

The same scaler is then applied to the test subjects. For the target, the training PHQ-9 values are first mapped to their observed ordinal indices. If $\mathcal{Y} = \{v_1 < \dots < v_K\}$ is the sorted set of PHQ-9 scores appearing in the training labels, then a score v_k is encoded as class index $k - 1$. We fit an ordinal ridge regressor [17] with $\alpha = 1.0$ to predict this ordered index:

$$f_{\text{ORIG}} : \text{StandardScaler} \rightarrow \text{OrdinalRidge}(\alpha = 1.0). \quad (6)$$

The predicted ordinal index is rounded internally by the model and mapped back to the corresponding PHQ-9 value in $\mathcal{Y}$. This output is used both as a continuous PHQ-9 predictor and as a confirmation signal in the final classification rule.

3.3 OpenFace Block

OpenFace is used as an interpretable facial behaviour extractor. In the preprocessed OpenFace arrays, the first column is the frame index; after removing it, the facial block keeps 43 columns:

$$8 \text{ gaze} + 17 \text{ AU intensities} + 18 \text{ AU presence indicators.} \quad (7)$$

For event $e \in \{1, 2, 3\}$, let $X^{(e)} \in \mathbb{R}^{T_e \times 43}$ denote the selected OpenFace sequence. We build four complementary summaries from this sequence, each trained as an independent PHQ-9 predictor.

AUVEL. The AUVEL view summarizes all three events. For each event, it concatenates the per-channel temporal mean, temporal

standard deviation, and mean absolute frame-to-frame change:

$$\phi_{\text{auvel}}^{(e)} = \left[\text{mean}_t X^{(e)}, \text{std}_t X^{(e)}, \frac{1}{T_e - 1} \sum_{t=1}^{T_e-1} |X_{t+1}^{(e)} - X_t^{(e)}| \right]. \quad (8)$$

Each event contributes $3 \times 43 = 129$ features, and concatenating the three events gives a 387-dimensional vector. The frame-change term is a discrete facial-motion proxy rather than a calibrated physical velocity.

E3ONLY. The E3ONLY view uses the same mean, standard deviation, and frame-change statistics, but only from event 3. It is 129-dimensional and is intended to emphasize the final-task facial state, which may reflect accumulated task effects.

Cross-event drift (FATIGUE). The FATIGUE code path represents cross-event drift. Let $\mu^{(e)} = \text{mean}_t X^{(e)}$. We encode

$$\phi_{\text{fatigue}} = [\mu^{(2)} - \mu^{(1)}, \mu^{(3)} - \mu^{(2)}, \mu^{(3)} - \mu^{(1)}], \quad (9)$$

which gives $3 \times 43 = 129$ features. This block captures whether expression, gaze, or AU activation changes systematically across the session.

AUSPEC. The AUSPEC view describes spectral facial dynamics. For each event, the 43 selected OpenFace channels are mean-centered and transformed with a real FFT. Let A_f be the magnitude spectrum for one channel. The frequency bins are split into low, middle, and high thirds. Following the submitted implementation, we compute three normalized spectral magnitude ratios, spectral entropy, and normalized dominant frequency:

$$p_f = \frac{A_f}{\sum_{f'} A_{f'} + \epsilon}, \quad (10)$$

with entropy and normalized dominant frequency

$$H = - \sum_f p_f \log(p_f + \epsilon), \quad d = \frac{\arg \max_f A_f}{F + \epsilon}. \quad (11)$$

Together with the three magnitude ratios, this gives five spectral statistics per channel, or $5 \times 43 = 215$ features per event. Concatenating the three events gives a 645-dimensional vector.

CCC-loss MLP training. Each facial view is standardized with a training-only StandardScaler and trained using the same two-hidden-layer MLP:

$$d \rightarrow 96 \rightarrow 48 \rightarrow 1, \quad (12)$$

with GELU activations and dropout rate 0.3. Since the leaderboard score contains CCC, the loss directly optimizes concordance with a small MSE term:

$$\mathcal{L} = (1 - \text{CCC}(y_m, \hat{y}_m)) + 0.1 \text{MSE}(y_m, \hat{y}_m). \quad (13)$$

The CCC term is computed as

$$\text{CCC}(y, \hat{y}) = \frac{2 \text{cov}(y, \hat{y})}{\text{var}(y) + \text{var}(\hat{y}) + (\text{mean}(y) - \text{mean}(\hat{y}))^2}. \quad (14)$$

Training uses mixup [23]. At each epoch, 2000 synthetic samples are drawn by sampling two training subjects a, b and a mixing coefficient $\lambda \sim \text{Beta}(0.4, 0.4)$:

$$X_m = \lambda X_a + (1 - \lambda) X_b, \quad y_m = \lambda y_a + (1 - \lambda) y_b. \quad (15)$$

One full-batch AdamW [15] update is then taken on these 2000 mixed samples. Each MLP is trained for 400 epochs with learning rate 10^{-3} , weight decay 10^{-3} , and gradient clipping at 1.0. Every 20 epochs, CCC is evaluated on the original training subjects, and the best checkpoint is retained. To reduce seed variance, each facial view is trained with 50 random seeds and the test predictions are averaged, producing f_{auvel} , f_{e3only} , f_{fatigue} , and f_{auspec} .

3.4 Fusion, Batch-Level Moment Alignment, and Classification

The final continuous PHQ-9 prediction is a fixed linear blend of the backbone and four OpenFace predictors:

$$\hat{y}_{\text{raw}} = 0.40f_{\text{orig}} + 0.30f_{\text{auvel}} + 0.20f_{\text{e3only}} + 0.10f_{\text{fatigue}} + 0.05f_{\text{auspec}}. \quad (16)$$

The weights are not constrained to sum to one, because the next step applies a global affine rescaling. They specify the relative contribution of the backbone and each facial view.

Small-sample regressors often shrink predictions toward the training mean. This is especially harmful for CCC, whose denominator penalizes not only errors in ranking but also mismatch in mean and variance, as shown in Eq. 14. We therefore apply a transductive, batch-level moment alignment step that maps the blended public-test predictions to target PHQ-9 moments:

$$\hat{y} = \text{clip} \left(\frac{\hat{y}_{\text{raw}} - \bar{\hat{y}}_{\text{raw}}}{\text{std}(\hat{y}_{\text{raw}}) + \epsilon} \sigma_t + \mu_t, 0, 27 \right). \quad (17)$$

Here $\bar{\hat{y}}_{\text{raw}}$ and $\text{std}(\hat{y}_{\text{raw}})$ are computed over the test batch, while the target moments are fixed to $\mu_t = 4.82$ and $\sigma_t = 3.92$ in the submitted configuration. This operation uses no individual test labels, but it does use the unlabelled test-batch distribution; it is therefore a transductive batch-level calibration step rather than a single-subject online calibration procedure. Before clipping, the alignment preserves the rank order of the blended predictions and corrects their location and scale. The clipping simply enforces the valid PHQ-9 range [0, 27].

The official task also evaluates binary and ternary labels. Instead of training a separate classifier, we derive labels from the calibrated PHQ-9 prediction and use the ORIG backbone as a confirmation branch. The binary prediction is

$$\hat{y}_2 = \mathbb{1}[\hat{y} \geq t_1 \wedge f_{\text{orig}} \geq 4]. \quad (18)$$

The ternary prediction is first thresholded from the calibrated score,

$$\hat{y}_3 = \begin{cases} 0, & \hat{y} < t_1, \\ 1, & t_1 \leq \hat{y} < t_2, \\ 2, & \hat{y} \geq t_2, \end{cases} \quad (19)$$

and is then forced to zero whenever $\hat{y}_2 = 0$. This conjunctive rule reduces false positives whose ensemble score is driven mainly by the facial predictors but is not supported by the gait-personality backbone. In the final submitted configuration, we use $t_1 = 4.25$ and $t_2 = 11.0$, selected from public leaderboard feedback.

4 Experiments

4.1 Experimental Protocol

The experiments are organized around the design choices of the final system and the Young modality settings. The G+P setting is represented by the IMU-personality ORIG backbone. Audio, ASR, and deep sequence trials are reported as non-final speech/video explorations. The final submitted system belongs to the A-V-G+P family, but its retained inputs are IMU acceleration, Big-Five scores, and OpenFace AU/gaze streams because the speech branches did not improve the recorded development results.

We report three types of empirical evidence. **Local design analyses** justify feature-cutting decisions and are not treated as official challenge scores. **Public-leaderboard checkpoints** document the cumulative development trajectory of the submitted system. **Non-final explorations** record speech and deep-sequence routes that were considered but excluded from the final ensemble. The official score combines the binary classification term, the PHQ-9 concordance term, and the ternary classification term:

$$\text{Score} = \frac{1}{3} (F1 + \text{CCC} + \text{Kappa}). \quad (20)$$

All leaderboard results are from submissions to the MPDD-AVG Young public split. No individual test labels were available; however, aggregate public-leaderboard feedback was used to select the final decision thresholds. We therefore treat the public split as a development benchmark rather than an unbiased held-out test set.

4.2 Effect of Compact Feature Design

We first examine whether the compact feature design in Sec. 3 is empirically justified. Table 1 reports local evidence for two final-system design choices: retaining only the acceleration channels from IMU and using compact OpenFace representations rather than raw high-dimensional descriptors.

For IMU, we isolate the channel-selection question by using IMU-only hand-crafted features and leave-one-out validation on the Young training set. Across the compared Ridge models, the 3-channel acceleration setting gives the lowest PHQ-9 MAE. Adding channels increases the feature dimension from 36 to 78, 129, and 189, but does not reduce the error. This supports the use of the first three acceleration channels as the IMU part of ORIG. For OpenFace, raw 710-dimensional features with a linear classifier perform at chance level in the local binary analysis, while PCA5 improves the binary accuracy to 0.682. A separate semantic subset analysis further shows that AU/gaze statistics outperform the complete semantic OpenFace feature set under the same direct binary evaluation. These results support the final facial design, which keeps AU/gaze streams and summarizes them through event-level, temporal-change, and spectral descriptors.

These analyses should be interpreted conservatively. They do not imply that lower dimensionality is always better; rather, they indicate that semantically constrained reductions are more reliable than broad high-dimensional inputs in this limited-sample setting.

Table 1: Local evidence for compact feature design. Panel A isolates IMU channel pruning with Young-train LOO Ridge regression; Panel B summarizes OpenFace dimensionality reduction and semantic feature selection with local binary classification. These numbers are design evidence, not official CodaBench scores.

A. IMU channel pruning (LOO Ridge, MAE ↓)			
Input channels	Dim	MAE	vs. 3-axis
3 accel. channels	36	3.444	–
6 channels	78	5.563	+2.119
9 channels	129	4.945	+1.501
12 channels	189	4.765	+1.321

B. OpenFace cutting (local LR, binary accuracy ↑)			
Comparison	Setting	Dim	Acc.
PCA compression	Raw descriptor	710	0.500
PCA compression	PCA5	5	0.682
Semantic subset	All semantic stats	479	0.682
Semantic subset	AU+gaze stats	399	0.727

Table 2: Historical public-leaderboard trajectory of the final system family. The rows are preserved cumulative development checkpoints, not strict leave-one-component-out ablations.

Checkpoint	Score	F1	CCC	Kappa
ORIG backbone	0.6701	0.7972	0.5083	0.7048
+ OpenFace AU/gaze	0.7063	0.7972	0.6170	0.7048
+ Frame-change / multi-view	0.7178	0.7972	0.6515	0.7048
+ Batch moment alignment	0.7931	0.7972	0.8772	0.7048
+ AND classification rule	0.8347	0.8361	0.8772	0.8012
+ Final thresholding	0.8745	–	0.8772	–

4.3 Cumulative Contribution of Final Components

Table 2 summarizes the preserved cumulative development checkpoints of the final system family on the public leaderboard. The table is not a strict leave-one-component-out ablation, because not every historical intermediate submission was preserved as an independent clean script. It is therefore used as historical public-leaderboard trajectory evidence rather than as a fully controlled ablation study.

The progression identifies changes that were associated with the main leaderboard increases. The ORIG backbone provides a low-dimensional baseline. Adding OpenFace AU/gaze features is associated with an increase in the continuous PHQ-9 component, with CCC moving from 0.5083 to 0.6170. Adding frame-change and multi-view facial predictors further moves CCC to 0.6515. The largest single change is batch-level moment alignment: after matching the prediction distribution to target PHQ-9 moments, CCC reaches 0.8772 and the overall score reaches 0.7931. Finally, the AND classification rule improves the discrete classification terms, raising F1 from 0.7972 to 0.8361 and Kappa from 0.7048 to 0.8012 while leaving the calibrated CCC unchanged in the preserved checkpoint.

The final submitted threshold configuration has a retained public Score of 0.8745 and CCC of 0.8772; its separate F1 and Kappa values were not preserved in the local result package.

4.4 Explored Routes Outside the Final Ensemble

We also evaluated routes that correspond to the broader available modalities but did not enter the final submission. Speech features included MFCC/openSMILE descriptors, wav2vec2 embeddings, temporal GRU models, and an ASR-based PHQ-9 symptom-prior route. These experiments were useful for model selection but were not competitive with the compact gait–personality and OpenFace pipeline: the GRU sequence models reached binary F1 values of 0.5856 for openSMILE and 0.5227 for wav2vec2 in recorded local experiments, and the standalone ASR route reached a public Score of 0.457751. We therefore report these audio and language routes as negative evidence under the 88-subject training regime, not as final-system components.

4.5 Final Submission and Reproducibility

The final submitted system uses three sources of information: IMU acceleration features, Big-Five personality scores, and OpenFace AU/gaze streams. The consolidated pipeline contains the ORIG backbone, the four facial predictors, linear fusion, batch-level moment alignment, and submission-file generation. It is preserved in the reproduction package through a single pipeline entry script, the ORIG builder, and the four OpenFace predictor builders, with standard scaling and linear models implemented through scikit-learn [16]. The final submitted configuration obtains a public leaderboard score of **0.8745**.

5 Limitations and Conclusion

We presented a compact multimodal system for the MPDD-AVG 2026 Young track. The system combines an IMU–personality ordinal backbone with four OpenFace AU/gaze temporal predictors, then applies linear fusion, transductive batch-level moment alignment, and a backbone-confirmed classification rule. The evidence supports the central design principle that, under limited supervision, semantically selected compact features and metric-aware post-processing can be competitive with broader high-dimensional modelling.

Several limitations should be made explicit. The public leaderboard was used as development feedback for decision thresholds, so the reported public score is not an unbiased estimate of hidden-test performance. The moment-alignment step uses the unlabelled test-batch distribution and is therefore not a single-subject online calibration procedure. The system is a challenge submission for depression-severity estimation, not a clinical diagnostic tool, and its behaviour across institutions, devices, and populations remains unverified. Within these conditions, the final submitted system achieves a public leaderboard Score of **0.8745**.

References

- [1] Alexei Baevski, Yuhao Zhou, Abdelrahman Mohamed, and Michael Auli. 2020. wav2vec 2.0: A framework for self-supervised learning of speech representations. *Advances in Neural Information Processing Systems* 33 (2020), 12449–12460.

- [2] Tadas Baltrušaitis, Chaitanya Ahuja, and Louis-Philippe Morency. 2019. Multimodal machine learning: A survey and taxonomy. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 41, 2 (2019), 423–443.
- [3] Tadas Baltrušaitis, Amir Zadeh, Yao Chong Lim, and Louis-Philippe Morency. 2018. OpenFace 2.0: Facial behavior analysis toolkit. In *IEEE International Conference on Automatic Face & Gesture Recognition (FG)*. 59–66.
- [4] Hanshu Cai, Zhenqin Yuan, Yiwen Gao, Shuting Sun, Na Li, et al. 2022. A multimodal open dataset for mental-disorder analysis. *Scientific Data* 9, 1 (2022), 178.
- [5] Jacob Cohen. 1960. A coefficient of agreement for nominal scales. *Educational and Psychological Measurement* 20, 1 (1960), 37–46.
- [6] Paul Ekman and Wallace V. Friesen. 1978. *Facial Action Coding System: A Technique for the Measurement of Facial Movement*. Consulting Psychologists Press, Palo Alto, CA.
- [7] Florian Eyben, Felix Weninger, Florian Gross, and Björn Schuller. 2013. Recent developments in openSMILE, the Munich open-source multimedia feature extractor. In *Proceedings of the 21st ACM International Conference on Multimedia*. 835–838.
- [8] Changzeng Fu, Zelin Fu, Qi Zhang, Xinhe Kuang, Jiacheng Dong, Kaifeng Su, Yikai Su, Wenbo Shi, Junfeng Yao, Yuliang Zhao, et al. 2025. The First MPDD Challenge: Multimodal Personality-aware Depression Detection. *arXiv preprint arXiv:2505.10034* (2025).
- [9] Changzeng Fu, Shiwen Zhao, Yunze Zhang, Zhongquan Jian, Shiqi Zhao, and Chaoran Liu. 2026. Personality-guided Public-Private Domain Disentangled Hypergraph-Former Network for Multimodal Depression Detection. In *Proceedings of the AAAI Conference on Artificial Intelligence*. arXiv:2511.12460.
- [10] Jonathan Gratch, Ron Artstein, Gale M Lucas, Giota Stratou, Stefan Scherer, et al. 2014. The distress analysis interview corpus of human and computer interviews. In *Proceedings of LREC*. 3123–3128.
- [11] Javier Si Zhao Hong, Timothy Zoe Delaya, Sherwyn Chan Yin Kit, Pai Chet Ng, and Xiaoxiao Miao. 2025. Exploring Machine Learning and Language Models for Multimodal Depression Detection. In *Asia Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC)*. arXiv:2508.20805.
- [12] Oliver P. John and Sanjay Srivastava. 1999. The Big Five trait taxonomy: History, measurement, and theoretical perspectives. In *Handbook of Personality: Theory and Research* (2 ed.), Lawrence A. Pervin and Oliver P. John (Eds.). Guilford Press, New York, NY, 102–138.
- [13] Kurt Kroenke, Robert L Spitzer, and Janet B W Williams. 2001. The PHQ-9: validity of a brief depression severity measure. *Journal of General Internal Medicine* 16, 9 (2001), 606–613.
- [14] Lawrence I-Kuei Lin. 1989. A concordance correlation coefficient to evaluate reproducibility. *Biometrics* 45, 1 (1989), 255–268.
- [15] Ilya Loshchilov and Frank Hutter. 2019. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*.
- [16] Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, et al. 2011. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research* 12 (2011), 2825–2830.
- [17] Fabian Pedregosa-Izquierdo. 2015. mord: Ordinal Regression in Python. <https://github.com/fabianp/mord>.
- [18] Fabien Ringeval, Björn Schuller, Michel Valstar, Nicholas Cummins, Roddy Cowie, et al. 2019. AVEC 2019 workshop and challenge: state-of-mind, detecting depression with AI, and cross-cultural affect recognition. In *Proceedings of the 9th International on Audio/Visual Emotion Challenge and Workshop*. 3–12.
- [19] Ying Shen, Huiyu Yang, and Lin Lin. 2022. Automatic depression detection: An emotional audio-textual corpus and a GRU/BiLSTM-based model. In *ICASSP 2022 - IEEE International Conference on Acoustics, Speech and Signal Processing*. 6247–6251.
- [20] Honghong Wang, Jing Deng, and Rong Zheng. 2025. Personality-Enhanced Multimodal Depression Detection in the Elderly. In *Proceedings of the 33rd ACM International Conference on Multimedia (MPDD Challenge)*. arXiv:2510.08004.
- [21] World Health Organization. 2017. Depression and other common mental disorders: global health estimates. World Health Organization.
- [22] Jeewoo Yoon, Chaewon Kang, Seungbae Kim, and Jinyoung Han. 2022. D-Vlog: Multimodal vlog dataset for depression detection. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 36. 12226–12234.
- [23] Hongyi Zhang, Moustapha Cisse, Yann N Dauphin, and David Lopez-Paz. 2018. mixup: Beyond empirical risk minimization. In *International Conference on Learning Representations (ICLR)*.